FIFTH SEMESTER

FC -1 ROCKET PROPULSION(AAE 3126)

30TH NOV 2024

Q1. Derive the equation of motion for a rocket. (2)

Ans: The equation of motion for a rocket is derived using the principles of **conservation of momentum** and **Newton's second law of motion**.

Step-by-Step Derivation:

1. **Momentum Conservation:**
   - At time t, the rocket's mass is $m(t)$ and its velocity is $v(t)$.
   - In a small time interval dt, the rocket expels a small amount of fuel dm (where $dm < 0$, as the mass decreases) with a velocity v_e relative to the rocket.
   - The change in momentum of the system (rocket + exhaust gases) must be zero, assuming no external forces.

2. **Momentum Before and After:**
   - Momentum before ejection:
   $$P_{\text{before}} = m(t)v(t)$$
   - Momentum after ejection:
   $$P_{\text{after}} = (m(t) + dm)(v(t) + dv) - dm(v(t) - v_e)$$
   Here, $v(t) - v_e$ is the absolute velocity of the exhaust gases.

4. **Applying Conservation of Momentum:**
   - Since $P_{\text{before}} = P_{\text{after}}$:
   $$m(t)v(t) = m(t)v(t) + m(t)dv + dm\,v_e$$

5. **Simplify the Equation:**
   - Cancel $m(t)v(t)$:
   $$0 = m(t)dv + dm\,v_e$$
   Rearrange to find:
   $$m(t)dv = -v_e\,dm$$

6. **Express Acceleration:**
   - The acceleration $a = \frac{dv}{dt}$:
   $$m(t)\frac{dv}{dt} = -v_e\frac{dm}{dt}$$
   This is the fundamental equation of motion for a rocket.

7. **Simplify Further:**
   - Let $\dot{m} = \frac{dm}{dt}$ (rate of change of mass, negative because the rocket is losing mass):
   $$m(t)\frac{dv}{dt} = v_e|\dot{m}|$$

Final Equation:

If we neglect external forces:

$$\frac{dv}{dt} = v_e \frac{|\dot{m}|}{m(t)}$$

If we include external forces F_{ext}, such as gravity or drag:

$$m(t)\frac{dv}{dt} = v_e|\dot{m}| + F_{\text{ext}}$$

Velocity (Integrating Over Time):

Integrating the simple form of the rocket equation gives the **Tsiolkovsky rocket equation**:

$$v(t) - v_0 = v_e \ln\left(\frac{m_0}{m(t)}\right)$$

where:

- $v(t)$ = velocity of the rocket at time t,
- v_0 = initial velocity,
- m_0 = initial mass,
- $m(t)$ = mass of the rocket at time t.

This equation relates the velocity of the rocket to the mass lost and exhaust velocity.

1m

Q2. Discuss any three key technical differences among solid, liquid, and hybrid propulsion systems. (3)

Ans: **Solid Propulsion**:

- Uses a single solid propellant (fuel and oxidizer are premixed and cast into the motor casing).
- Easy to store and handle because the propellant is stable and inert until ignited.
- Cannot be throttled or stopped once ignited; burns until all the propellant is consumed.

1m

Liquid Propulsion:

- Uses liquid fuel and liquid oxidizer stored in separate tanks, requiring complex plumbing and pumps to deliver them to the combustion chamber.
- Offers precise control: can be throttled, shut down, and restarted during flight.
- Requires cryogenic or high-pressure storage, making it less convenient to handle and store.

1m

Hybrid Propulsion:

- Combines a solid fuel and a liquid (or gaseous) oxidizer stored separately.
- Easier to handle than liquid systems (solid fuel is inert until oxidizer is introduced).
- Can be throttled and shut down, though less efficiently than liquid systems.

1m

Summary:

Feature	Solid Propulsion	Liquid Propulsion	Hybrid Propulsion
Propellant State	Solid	Liquid	Solid fuel + liquid oxidizer
Throttling	Not possible	Fully controllable	Limited control
Design Complexity	Simple	Highly complex	Moderate
Efficiency	Moderate	High	Between solid and liquid

Q3. An end-burning rocket uses a cylindrical double base propellant grain with a diameter of 200 mm and generates a thrust of 350 N over a period of 300 s. The thrust coefficient is 1.15. The characteristics of the propellant are; Density of propellant grain =1500 kg/m^3, Speed of Sound (a_{70})=4 mm/s, Choice of index (n)=0.5, Characteristic velocity (C*)=1500 m/s.

Determine the thrust developed by the solid propellant rocket given in the above example, when the rocket is operated during daytime in a hot desert, wherein the ambient temperature is 50^{0}C and the propellant grain is soaked to the above temperature. The design of them can be assumed to be for a nominal initial propellant temperature of 25^{0}C. The values of C* and C_F can be assumed not to change with the initial temperature of the grain. The temperature sensitivity coefficient is 0.0035^{0}C^{-1}.(5)

Sol:

Solution: Temperature sensitivity coefficient $\pi_r = \frac{d(\ln r)}{dT} = 0.0035$ 1m

The change in burn rate with temperature is given by $\frac{r_2}{r_1} = e^{\pi_r (T_2 - T_1)}$, giving the burn rate at 50°C as $1.512 \times 10^{-6} \times p^{0.5}\, e^{.0035(50-25)}$ m/s $= 1.650 \times 10^{-6} \times p^{0.5}$ m/s 1m

The equilibrium chamber pressure is obtained from the balance between mass generated and mass flow rate through the nozzle and is:

$$p = \left(\frac{S_b \rho_p a C^*}{A_t} \right)^{\frac{1}{1-n}} = \left(\frac{0.0314 \times 1500 \times 1.650 \times 10^{-6} \times 1500}{37.63 \times 10^{-6}} \right)^{\frac{1}{0.5}} = 9.6 \times 10^6 \text{ Pa.}$$ 1m

This is higher than the chamber pressure of 8.09×10^6 Pa at 25°C. The thrust would, therefore, be higher and is given by $F = C_F\, p\, A_t = 1.15 \times 9.60 \times 10^6 \times 37.63 \times 10^{-6} = 415.4$ N. 1m

The thrust has gone up from 350 N to 415.4 N, i.e. by about 18.7%. 1m

Q4. What is rocket staging? Discuss its advantages and limitations. (2)

Ans: **Rocket staging** is the practice of using multiple connected sections, or **stages**, in a rocket to sequentially jettison parts of the vehicle as they become unnecessary during flight. Each stage has its own engines and propellant, and stages are discarded once their fuel is depleted. 1m

This process allows the rocket to shed excess weight and increase efficiency as it ascends, making it possible to reach higher velocities and altitudes.

Advantages of Rocket Staging:

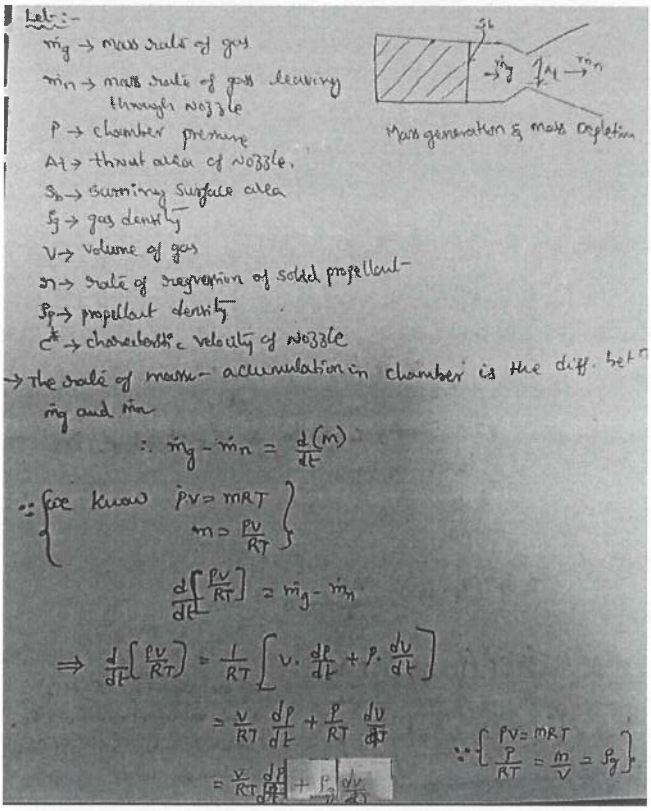
1. **Increased Efficiency**:
2. **Extended Range and Payload Capacity**: ½ m
3. **Modularity**:
4. **Flexibility**:

Limitations of Rocket Staging: ½ m

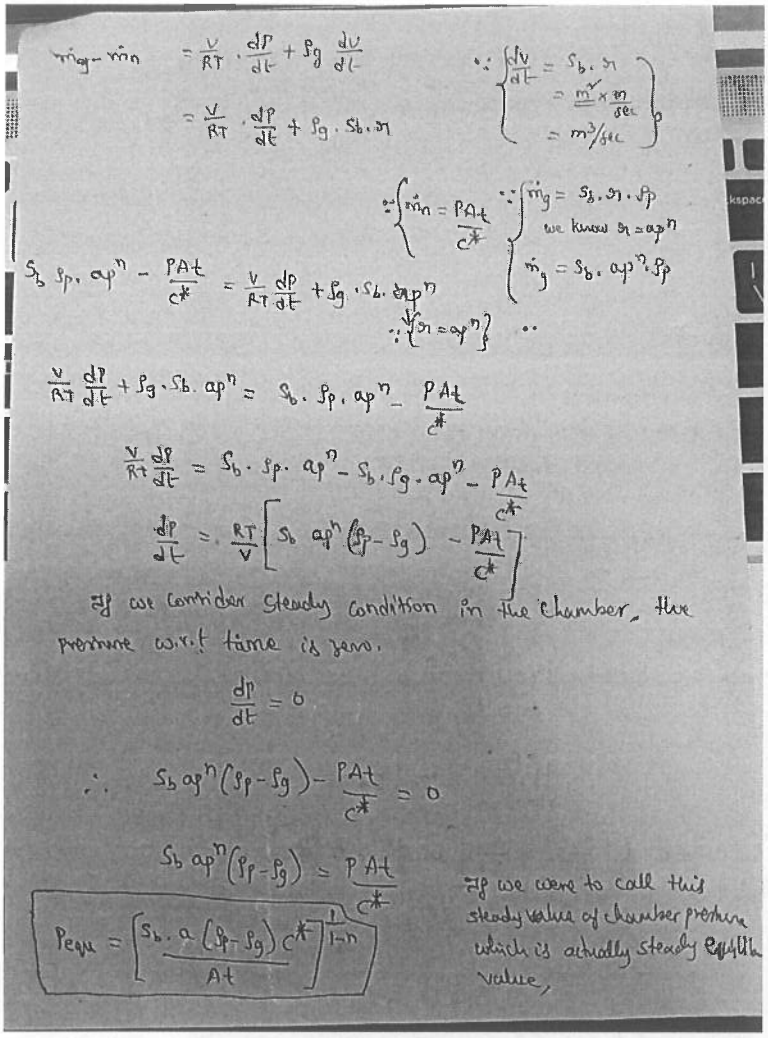
1. **Complexity**:
2. **Cost**:
3. **Weight Penalty for Structural Components**:
4. **Efficiency Loss During Staging**:
5. **Environmental Impact**:

Q5. Derive the expression for equilibrium pressure (P_{equ}) for the stable operation of a solid propellant rocket. (3)

Ans:



1½ m



Q6. A four stage rocket is used to put up a satellite of 40 kg mass in a Low Earth Orbit (LEO). The approximate values of mass of the propellant of mass of the propellant, mass of structure and jet velocity for each stage are given below:

STAGE	I	II	III	IV
Mass of Propellant (kg)	9000	3500	1700	260
Mass of structure including inerts, (kg)	1500	550	250	40
V_j (m/s)	2200	2400	2500	2750

Determine: (i) The payload mass fraction of the total rocket (satellite launch vehicle). (ii) Structural mass fraction of each stage (iii) The ideal incremental velocity by each stage and the total incremental velocity. (iv) If the first stage fires for a period of 50 seconds and the rate of mass depletion can be assumed to be constant, what would be the acceleration of the rocket at the take-off?. (5)

Solutions:

Solution: (a) Payload mass fraction of the satellite launch vehicle

$$= \frac{40}{(9000+3500+1700+260)+(1500+550+250+40)+40}$$

$$= \frac{40}{16840} = 2.37\times10^{-3}$$

1m

(b) Structural mass fractions:

First stage: $\frac{1500}{1500+9000} = 0.143$

Second stage: $\frac{550}{550+3500} = 0.136$

Third stage: $\frac{250}{250+1700} = 0.128$

Fourth stage: $\frac{40}{40+260} = 0.133$

1m

(c) Ideal ΔV provided by each stage:

First stage:

$$2200\ln\frac{(9000+3500+1700+260)+(1500+550+250+40)+40}{(3500+1700+260)+(1500+550+250+40)+40} = 1.682 \text{ km/s}$$

Second stage:

$$2400\ln\frac{(3500+1700+260)+(550+250+40)+40}{(1700+260)+(550+250+40)+40} = 1.927 \text{ km/s}$$

Third stage:

$$2500\ln\frac{(1700+260)+(250+40)+40}{(260)+(250+40)+40} = 3.39 \text{ km/s}$$

1m

Fourth stage:

$$2750\ln\frac{(260)+(40)+40}{(40)+40} = 3.979 \text{ km/s}$$

Total $\Delta V = 1.682 + 1.927 + 3.39 + 3.979 = 10.978$ km/s

d) Since the first stage fires for a period of 50 second and the rate of propellant mass depletion is constant, the mass flow rate of propellant $\dot{m} = \frac{\text{mass of propellant}}{\text{time}} = \frac{9,000}{50} = 180$ kg/s.

1m

Thrust, F, of the stage $= \dot{m}V_j = 180\times2200$ N

Total mass at lift off $= M_i = (9000+3500+1700+260)+(1500+550+250+40)+40$

$= 16,840$ kg

Initial acceleration $= \frac{F-M_i g}{M_i} = \frac{180\times2200-16840\times9.81}{16840} = 13.705 \text{ m/s}^2$

In the above equation the acceleration due to gravity, g, is 9.81 m/s^2.

Hence, net take off acceleration is 13.705 m/s^2 which is 1.4 times the acceleration due to gravity, i.e. 1.4 g.

1m

Q7. What is combustion instability? Describe any two methods to prevent combustion instability in rockets. (2)

Ans: **Combustion instability** refers to oscillations or fluctuations in the pressure, temperature, and flow of the combustion process within a rocket engine. These instabilities can range from minor fluctuations to destructive resonant oscillations that can damage or destroy the engine.

1m

Combustion instability is a critical challenge in rocket engine design that can lead to catastrophic failure if not managed effectively. Methods like **baffles and acoustic dampers** and **optimized propellant injection** are essential tools to ensure stable, reliable performance in rocket propulsion systems.

1m

Q8. The altitude and orbit of a satellite are maintained using a number of small rockets housed in the satellite. The altitude and orbit corrections required during the lifetime of satellite are estimated to be 950 m/s. if the jet velocity of rocket is 2500 m/s and dry mass of satellite (Dry mass without the propellant being loaded in the satellite) is 800 kg, determine the mass of propellant required for the altitude and orbit corrections. (3)

Sol:

Sol:

Given $V_j = 2500$ m/s

$\Delta V = 950$ m/s

$M_f = 800$

$\Delta V = V_j \cdot \ln\left(\frac{M_i}{M_f}\right) \Rightarrow 950 = 2500 \cdot \ln\left(\frac{800 + m_p}{800}\right)$ 1m

$m_p = 370$ kg. 1m

Q9. A high pressure-fed liquid propellant rocket based on the gas generator cycle has a vacuum thrust of 735 KN and burn duration of 180 s. The propellants used are Nitrogen Tetroxide (N_2O_4) and Unsymmetrical Dimethyl Hydrogen (UDMH). The specific impulse is 2950 N-s/kg. The Mixture ratio is 1.87. The pressure in the thrust chamber of the rocket is 6 MPa and the propellant supply pressure to the chamber is 7 MPa. N_2O_4 is stored in the propellant tank at a pressure of 0.4 MPa and UDMH is stored at 0.32 MPa. The densities of N_2O_4 and UDMH are 1400 Kg/m^3 and 790 Kg/m^3 respectively, at the temp used in the rocket. Determine power required driving N_2O_4 and UDMH pump. (5)

Sol:

Solution: A simplified schematic diagram is illustrated in Fig. 6.44.

(*i*) Power required for driving the pumps:

Thrust $F = 735$ kN

Specific Impulse $I_{sp} = 2950$ Ns/kg

Propellant mass flow rate $\dot{m}_p = \frac{F}{I_{sp}} = \frac{735000}{2950} = 249.15$ kg/s

Mixture ratio $R = 1.87$

Fuel (UDMH) mass flow rate $\dot{m}_U = \frac{\dot{m}_p}{1+R} = \frac{249.15}{2.87} = 86.81$ kg/s

Oxidizer (N_2O_4) mass flow rate $\dot{m}_N = \frac{R\dot{m}_p}{1+R} = \frac{1.87 \times 249.15}{2.87} = 162.34$ kg/s

3m

Volumetric flow rate of UDMH $\dot{Q}_U = \frac{\dot{m}_U}{\rho_U} = \frac{86.81}{790} = 0.11 \frac{m^3}{s}$

Volumetric flow rate of N_2O_4 $\dot{Q}_N = \frac{\dot{m}_N}{\rho_N} = \frac{162.34}{1400} = 0.115$ m^3/s

Pressure rise across UDMH pump $\Delta p_U = 7 - 0.32 = 6.68$ MPa
Pressure rise across N_2O_4 pump $\Delta p_N = 7 - 0.4 = 6.6$ MPa

1m

Power of UDMH pump $= \dot{Q}_U \Delta p_U = 0.11 \times 6.68 \times 10^6$ W $= 0.735 \times 10^6$ W
Power of N_2O_4 pump $= \dot{Q}_N \Delta p_N = 0.115 \times 6.6 \times 10^6$ W $= 0.765 \times 10^6$ W

1m

Q10. Define specific impulse and impulse, along with their units. (2)

Ans: **Impulse** and **specific impulse** are fundamental concepts in rocketry and propulsion systems, used to measure a rocket engine's performance.

Impulse

1. **Definition:**
   - Impulse is the total momentum change produced by a rocket engine over a given time.
   - Mathematically, impulse (I) is given by:

$$I = F \cdot \Delta t$$

   where:
   - F: Thrust (force produced by the engine, in newtons, N),
   - Δt: Time duration over which the thrust is applied (in seconds, s).

1m

2. **Units:**
   - The unit of impulse is newton-seconds (N·s) in the SI system.

Specific Impulse (I_{sp})

1. **Definition:**
   - Specific impulse is a measure of the efficiency of a rocket engine.
   - It is defined as the total impulse produced per unit of propellant weight.
   - Mathematically:

$$I_{sp} = \frac{I}{m \cdot g_0} = \frac{F}{\dot{m} \cdot g_0}$$

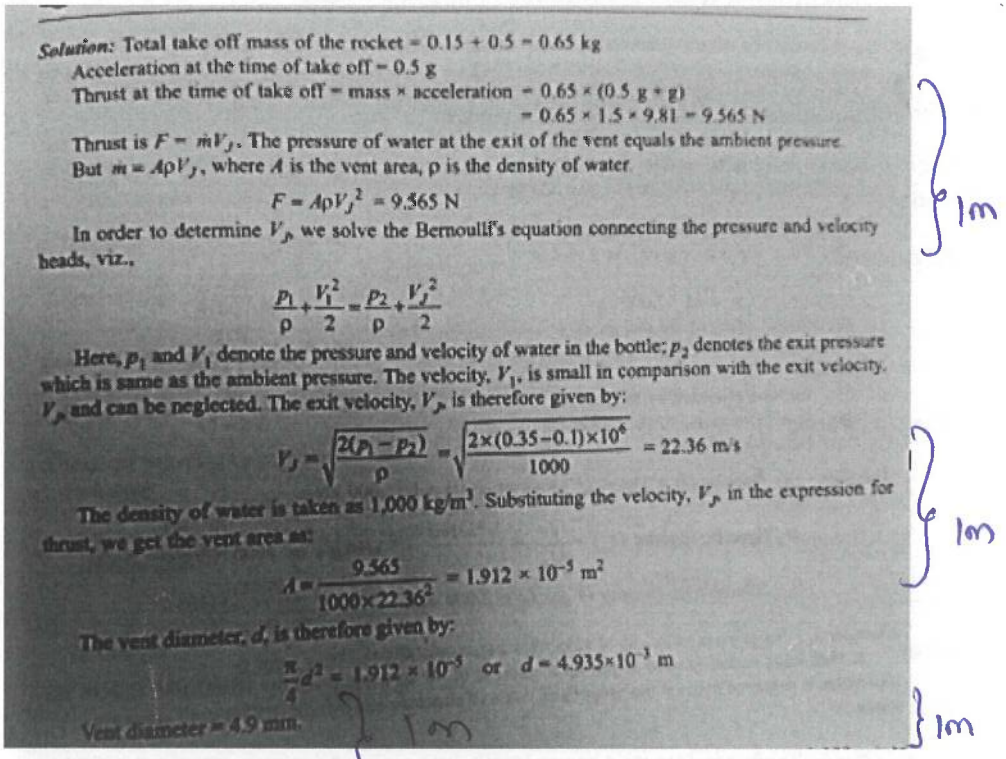
   where:
   - I: Total impulse (N·s),
   - m: Mass of the propellant (kg),
   - $\dot{m}$: Propellant mass flow rate (kg/s),
   - g_0: Standard gravitational acceleration ($9.81\ \mathrm{m/s^2}$).

1m

2. **Units:**
   - **Seconds (s)** in SI units (as it represents the time for which one unit weight of propellant can produce thrust).

Q11. A small water rocket is made by pressurising a plastic bottle partially filled with water using compressed air at 0.35 MPa at the ambient temperature and releasing the pressurised water through a vent. The mass of water in the rocket is 0.5 kg. The mass of the empty bottle, which constitutes the mass of the structure and inert components of the rocket, is 0.15 kg. If the useful payload is zero and the ambient pressure is 0.1 MPa, determine the vent diameter required for vertical takeoff with an initial acceleration of 0.5 g. Neglect the pressure from the height of water in the bottle. (3)

Sol:



Solution: Total take off mass of the rocket = 0.15 + 0.5 = 0.65 kg

Acceleration at the time of take off = 0.5 g

Thrust at the time of take off = mass × acceleration = 0.65 × (0.5 g + g)

= 0.65 × 1.5 × 9.81 = 9.565 N

Thrust is $F = \dot{m}V_j$. The pressure of water at the exit of the vent equals the ambient pressure.

But $\dot{m} = A\rho V_j$, where A is the vent area, ρ is the density of water.

$$F = A\rho V_j^2 = 9.565 \text{ N}$$

In order to determine V_j, we solve the Bernoulli's equation connecting the pressure and velocity heads, viz.,

$$\frac{p_1}{\rho} + \frac{V_1^2}{2} = \frac{p_2}{\rho} + \frac{V_j^2}{2}$$

Here, p_1 and V_1 denote the pressure and velocity of water in the bottle; p_2 denotes the exit pressure which is same as the ambient pressure. The velocity, V_1, is small in comparison with the exit velocity, V_j, and can be neglected. The exit velocity, V_j, is therefore given by:

$$V_j = \sqrt{\frac{2(p_1 - p_2)}{\rho}} = \sqrt{\frac{2\times(0.35-0.1)\times10^6}{1000}} = 22.36 \text{ m/s}$$

The density of water is taken as 1,000 kg/m^3. Substituting the velocity, V_j, in the expression for thrust, we get the vent area as:

$$A = \frac{9.565}{1000\times22.36^2} = 1.912 \times 10^{-5} \text{ m}^2$$

The vent diameter, d, is therefore given by:

$$\frac{\pi}{4}d^2 = 1.912 \times 10^{-5} \quad \text{or} \quad d = 4.935\times10^{-3} \text{ m}$$

Vent diameter = 4.9 mm.

1m

1m

1m

1m

Q12. Explain the different types of injectors used in liquid propellant rockets. (5)

Ans: Injectors in liquid-propellant rockets play a crucial role in delivering and mixing the fuel and oxidizer in the combustion chamber for efficient and stable combustion. The type of injector directly influences combustion efficiency, thrust stability, and prevention of combustion instabilities.

Types of Injectors in Liquid Propellant Rockets

1. **Showerhead Injector**: 1m
   - **Description**:
     - A simple design where fuel and oxidizer are injected through a grid of small holes, resembling a showerhead.
     - The streams are directed into the combustion chamber.
   - **Advantages**:
     - Easy to manufacture and reliable.
   - **Disadvantages**:
     - Limited mixing efficiency, leading to suboptimal combustion.

- Susceptible to combustion instabilities.

---

2. **Impinging Injector**: \ m

   - **Description**:
     - Fuel and oxidizer streams are directed at each other at specific angles to collide (impinge).
     - The collision causes atomization, improving mixing.
   - **Types**:

1. **Double Impinging**: Fuel and oxidizer streams collide directly.

2. **Self-Impinging**: Streams of the same propellant (e.g., oxidizer) collide and mix with the other propellant after impact.

   - **Advantages**:
     - High atomization efficiency.
     - Effective for small and medium engines.
   - **Disadvantages**:
     - Complex design requiring precise alignment.
     - Potential for clogging or wear at impingement points.

---

3. **Swirl Injector**: \ m

   - **Description**:
     - Propellants are injected tangentially into swirl chambers, creating a vortex.
     - The swirling motion atomizes the liquid into fine droplets before exiting through the nozzle.
   - **Advantages**:
     - Excellent mixing and atomization.
     - High combustion efficiency.
   - **Disadvantages**:
     - Higher manufacturing complexity and cost.
     - Increased risk of cavitation in the propellant feed lines.

---

4. **Pintle Injector**: \ m

   - **Description**:
     - Features a movable pintle (a rod or plug) inside a coaxial injector. The propellants are injected in a concentric, adjustable pattern.

- Often used in throttling engines.

- **Advantages**:
  - Simple design, robust, and highly reliable.
  - Good mixing efficiency and inherent combustion stability.
  - Easily adaptable for throttling.
- **Disadvantages**:
  - Limited scalability for very large engines.
  - Asymmetric flow patterns can occur at high throttling levels.

---

5. **Like-on-Like Injector**: 1m
   - **Description**:
     - Both fuel and oxidizer are injected in parallel streams without impingement.
     - Mixing occurs primarily through diffusion and turbulence in the combustion chamber.
   - **Advantages**:
     - Simplifies injector design and reduces mechanical complexity.
   - **Disadvantages**:
     - Lower mixing efficiency.
     - More prone to incomplete combustion.

---

6. **Coaxial Injector**: ½ 1m
   - **Description**:
     - Fuel and oxidizer are injected in concentric streams. For example, oxidizer flows through a central core, and fuel flows in an annular stream around it.
     - This design promotes shear mixing between the layers.
   - **Advantages**:
     - Effective mixing with good control over injection flow rates.
     - High stability, commonly used in cryogenic engines (e.g., liquid oxygen and hydrogen engines).
   - **Disadvantages**:
     - Requires precise manufacturing tolerances.
     - Complex design increases cost.

---

7. **Gas-Liquid Injector**: ½ 1m

- **Description**:
  - One propellant is injected as a gas, and the other as a liquid. The gas helps atomize the liquid upon interaction.
- **Advantages**:
  - Improves atomization and mixing.
  - Useful in engines where one propellant can be preheated to gaseous form (e.g., gas-generator cycles).
- **Disadvantages**:
  - Requires pre-heating systems for one of the propellants.
  - Higher design and operational complexity.

Q13. Derive an expression for the residence time in a rocket. (2)

Ans: **Residence time** in a rocket's combustion chamber is the average time a propellant particle spends in the chamber before being expelled through the nozzle. This is a crucial parameter for ensuring proper mixing and combustion of propellants.

Derivation of the Residence Time Expression

1. **Definition of Residence Time**: The residence time (t_r) is given by:

$$t_r = \frac{\text{Volume of the combustion chamber}}{\text{Volumetric flow rate of propellant through the chamber}}$$

2. **Combustion Chamber Volume**: Let V_c represent the volume of the combustion chamber (in m^3).
3. **Volumetric Flow Rate**:
   - The volumetric flow rate ($\dot{V}$) is defined as the mass flow rate of the propellant ($\dot{m}$) divided by its density (ρ):

$$\dot{V} = \frac{\dot{m}}{\rho}$$

4. **Expression for Residence Time**: Substituting $\dot{V}$ into the definition of residence time:

$$t_r = \frac{V_c}{\dot{V}} = \frac{V_c}{\frac{\dot{m}}{\rho}}$$

Simplify:

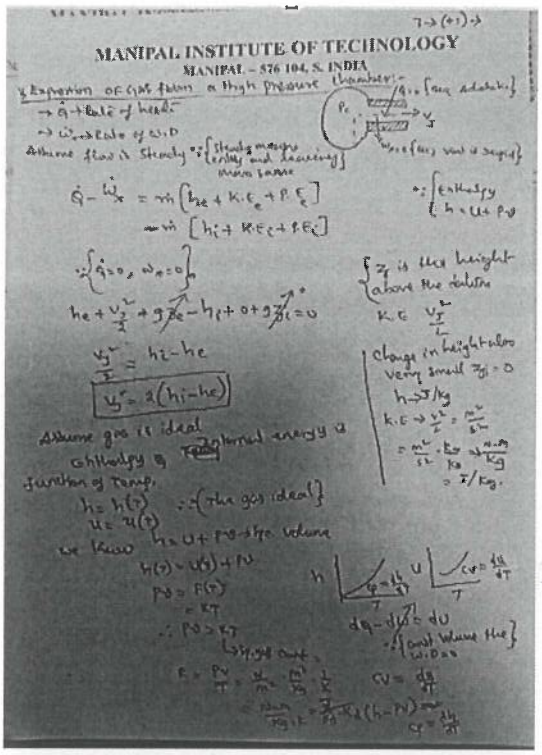
$$t_r = \frac{V_c \cdot \rho}{\dot{m}}$$

1m

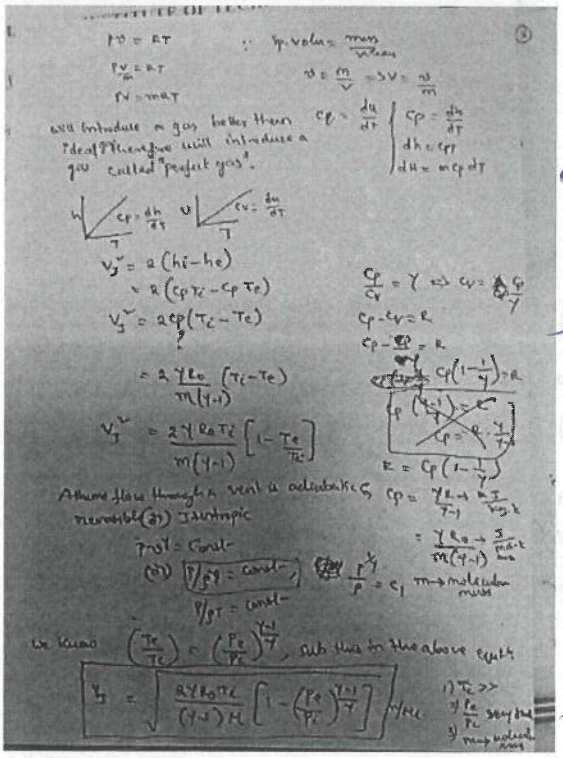
1m

Q14. Derive an expression for the ejection velocity "V_j" of gas from a high-pressure chamber and draw the conclusions. (3)

Sol:



1m



1m

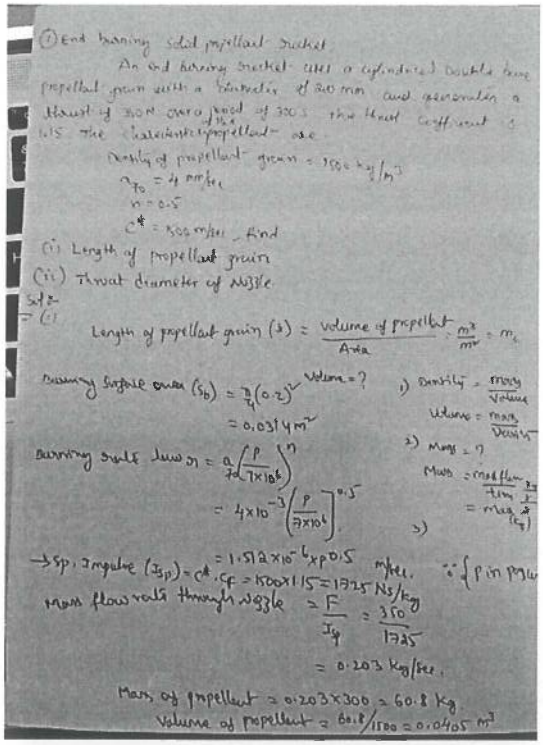
1m

Q15. An end burning rocket uses a cylindrical double base propellant grain with a diameter of 200 mm and generates a thrust of 350 N over a period of 300 s. the thrust coefficient is 1.15. The characteristics of the propellants are: Density of propellant grain = 1500 kg/m^3. Speed of Sound(a_{70}=4mm/s).

Choice of index (n)= 0.5

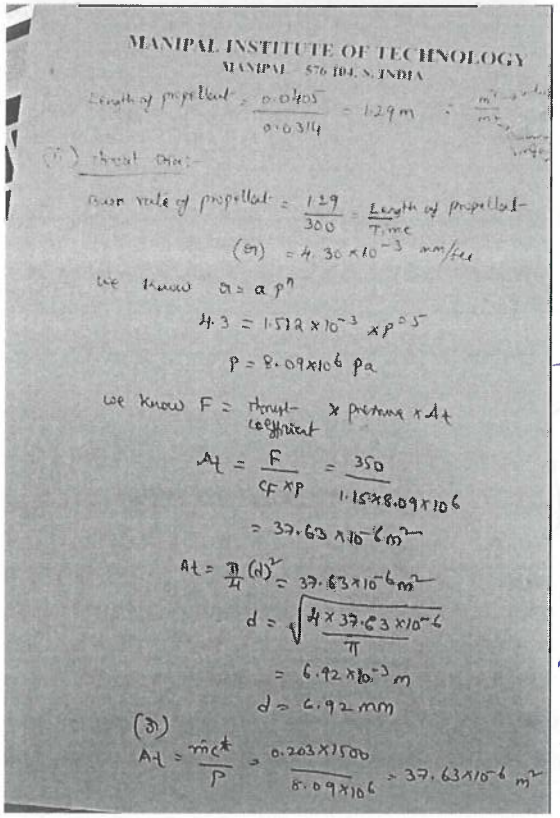
Characteristic velocity (C*) = 1500 m/sec . Calculate (i) Length of propellant grain , and (ii)Throat diameter of the nozzle. (5)

Sol:



1m

1m



1m

1m

1m